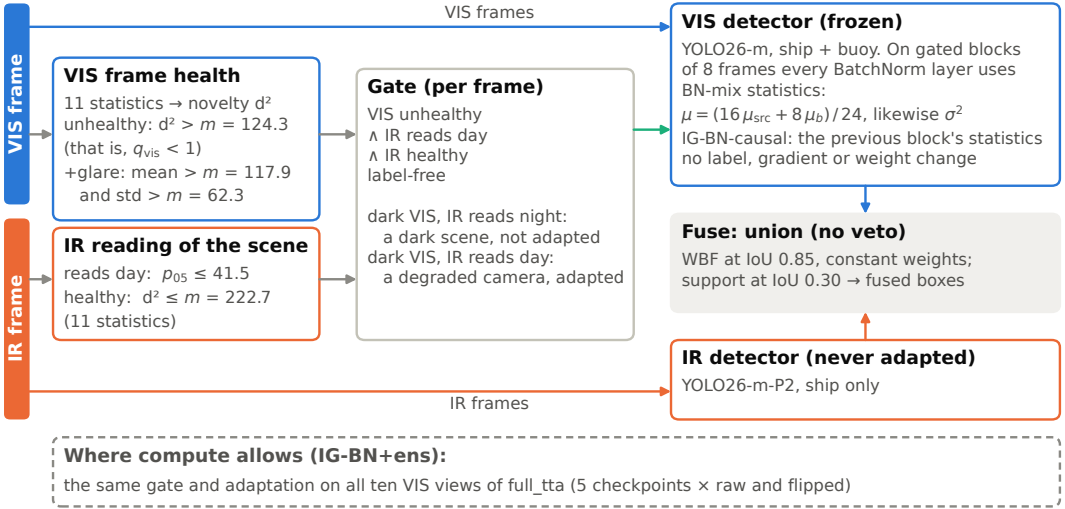
VIS frame
IR frame
VIS frames
VIS frame health
11 statistics → novelty d²
unhealthy: d² > m = 124.3
(that is, q_vis < 1)
+glare: mean > m = 117.9
and std > m = 62.3
IR reading of the scene
reads day: p05 ≤ 41.5
healthy: d² ≤ m = 222.7
(11 statistics)
Gate (per frame)
VIS unhealthy
∧ IR reads day
∧ IR healthy
label-free
dark VIS, IR reads night:
a dark scene, not adapted
dark VIS, IR reads day:
a degraded camera, adapted
VIS detector (frozen)
YOLO26-m, ship + buoy. On gated blocks of 8 frames every BatchNorm layer uses BN-mix statistics:
μ = (16 μ_src + 8 μ_b) / 24, likewise σ²
IG-BN-causal: the previous block's statistics no label, gradient or weight change
Fuse: union (no veto)
WBF at IoU 0.85, constant weights; support at IoU 0.30 → fused boxes
IR detector (never adapted)
YOLO26-m-P2, ship only
IR frames
Where compute allows (IG-BN+ens):
the same gate and adaptation on all ten VIS views of full_tta (5 checkpoints × raw and flipped)